

Pixel Art Is a Palette, Not a Downscale: Rebuilding Targets and Decoding for Tiny Sprite Generation

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Abstract

Text-to-image models fail at the smallest images people draw on purpose: 12–24px RGBA game sprites, where a handful of colours and hard alpha edges are the medium. We show that the usual failure is not a modelling failure but a target failure. In the sprite corpus this area trains on, only 7.1% of the images are natively 16 px or smaller; the rest is 25–64px artwork box-downscaled into the low-resolution buckets, and 4,888 pixel-identical duplicates leave a twin in training for 16.2% of the held-out set. A model fitted to those targets learns soft, fifty-colour blobs, and the evaluation reference, drawn from the same pool, rewards exactly that. We rebuild the targets and evaluate against references that contain only real low-resolution pixel art. Under this protocol the data work alone takes FD-DINOv2 from 167.2 to 117.9 at 16px. We then close the remaining gap where it actually lies: real sprites use a median of six opaque colours while every sampling rule we tested emits more than fifty. Projecting the predicted $\hat{x}_0$ onto a per-sprite k -colour palette during the late denoising steps reaches 31.8 ± 0.5 — better than quantising finished samples (41.6), than one corpus-wide palette (120.1) and than a median filter that removes the same dither (112.2). The rule is a property of the recipe rather than of the network: a transformer backbone that trails the UNet by 36.5 FD without it lands within 0.7 with it. Applied to sprites from generators we did not train, it improves native FD by 36–41% while preserving 80% of the silhouette. Against external systems given the same palette post-process we reach 50.8 versus 134.4 for gpt-image-2 with downscaling.

1. Introduction

Game sprites are among the smallest images anyone draws on purpose: an item or a character on a 12–24px



Figure 1. Four rows of 16px RGBA sprites, nearest-neighbour magnified, with the median number of opaque colours per row. Real sprites drawn at this size use six; the corpus’s own 16px targets, which are 25–64px artwork box-downscaled, use 49; gpt-image-2 rendered large and reduced uses 111; ours uses four. The third row is the easiest on the eye and the furthest from the medium.

RGBA canvas, built from hard edges, a handful of flat colour regions and a transparent background. At these sizes there is no texture in which to hide a mistake — one wrong pixel changes the silhouette, one blended colour reads as blur. The obvious recipe, prompting a large text-to-image model and shrinking its output, produces something recognisable but not a sprite: it keeps the soft shading and the hundred-odd colours of an illustration (Figure 1).

This paper is about a failure one level below the model. We train a dedicated generator for this medium — a 72.5M pixel-space diffusion model over RGBA sprites, conditioned on a frozen CLIP text encoder and a resolution-bucket label — and find its weaknesses inherited from what it is asked to imitate. In the sprite corpus this area trains on, only 7.1% of the images are natively 16px or smaller; two thirds are 25–64px artwork that the pipeline box-downsamples into the low-resolution buckets, and 4,888 pixel-identical duplicates leave a twin in training for 16.2% of the held-out set. A model fitted to those targets learns what they are: soft, fifty-colour blobs. Worse, the standard evaluation

058 reference is drawn from the same pool, so it rewards
059 the blur it caused, and the ranking of methods under
060 it does not survive contact with real pixel art.

061 We change both the targets and the yardstick. On
062 the data side we remove the duplicates, forbid feeding a
063 sprite to a bucket more than $1.5\times$ below its native size,
064 and add 14,999 natively small sprites from permissively
065 licensed collections, captioned with a vision–language
066 model. On the evaluation side we define the native
067 protocol: FD-DINOv2 against real sprites that are na-
068 tively at most R px, never downsampled. At 16 px the
069 data work alone moves the model from 167.2 to 117.9.

070 What remains is a property of the medium that con-
071 tinuous diffusion cannot express. Real 16 px sprites
072 use a median of six opaque colours and 46% of their
073 adjacent opaque pixel pairs are exactly equal; every
074 sampling rule we tested emits more than fifty colours
075 and reaches 8%, spending the difference on few-level
076 dither inside regions that should be flat. We close that
077 gap inside the sampler: during the late denoising steps
078 we project the predicted $\hat{x}_0$ onto a per-sprite k -colour
079 palette (batched k -means over the opaque pixels, al-
080 pha untouched) and continue, so the remaining steps
081 repair the seams hard quantisation leaves. At 16 px
082 this takes FD from 87.1 to 31.8 ± 0.5 , where quantising
083 the finished samples reaches only 41.6.

084 Two results say the rule is not an artefact of our net-
085 work. Replacing the UNet with a 69.7M transformer,
086 everything else held fixed, gives a model that trails by
087 36.5 FD under plain guidance and by 0.7 once the pro-
088 jection is on: the decoding rule absorbs the difference
089 between the two architectures. And the rule applies to
090 sprites we did not generate — re-noising the output of
091 gpt-image-2, FLUX.2 with a pixel-art LoRA, or SDXL
092 with Pixel Art XL and denoising it under our recipe im-
093 proves native FD by 36–41% at every resolution while
094 keeping about 80% of the input silhouette.

095 We also report a negative result we consider as in-
096 formative as the positive one. Cross-resolution self-
097 guidance was, under the downscale-biased reference,
098 our strongest method by 44%; under the native pro-
099 tocol it gives nothing. Which reference set one scores
100 against decides which of these conclusions one pub-
101 lishes.

102 Contributions.

- 103 • A diagnosis of the targets used for tiny-sprite gener-
104 ation — native-resolution statistics, duplicate leak-
105 age, and the resulting bias of the standard reference
106 — and a rebuilt corpus plus the native protocol that
107 removes both.
- 108 • Palette-projected sampling, which lowers FD-
109 DINOv2 at 16 px from 87.1 to 31.8 ± 0.5 over three

seeds, beats post-hoc quantisation (41.6) and two
controls that isolate what it is doing, and costs 0.8
points of caption retrieval at $k=6$, where the samples
match the colour statistics of real sprites.

- Evidence that the rule belongs to the recipe rather
than the network: it absorbs 98% of the gap be-
tween two architectures, and it improves the sprites
of three systems we did not train.
- An external comparison in which every system re-
ceives the same palette post-process: at 16 px we
reach 50.8 against 134.4 for gpt-image-2 with down-
scaling, 150.1 for FLUX.2 with a pixel-art LoRA and
244.2 for SDXL.

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